

Antibiotic Resistance Profiles and the Pathogenicity of Medical Devices-Associated Bacterial Strains

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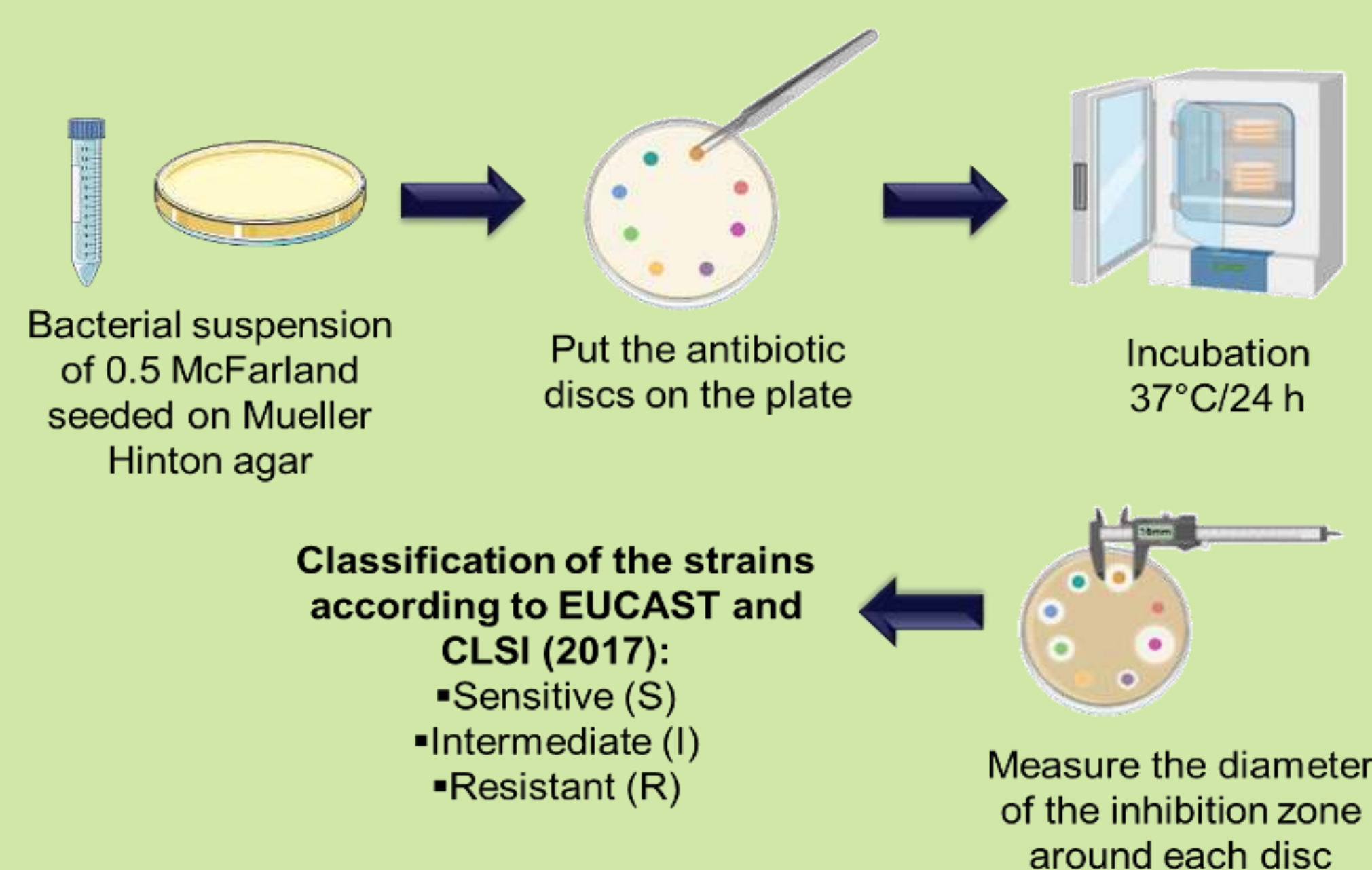
Introduction

The failure of antibiotic therapy to treat infections has become a major global public health concern in recent years (Levin-Reisman *et al.*, 2019; de la Fuente-Nunez *et al.*, 2023). Biofilm development is another important factor that contributes to treatment failure in addition to antibiotic resistance (de la Fuente-Nunez *et al.*, 2023). Because biofilm-embedded bacteria are more resistant to antibiotics, they can withstand antimicrobial therapy and grow again (Usui *et al.*, 2023). The recalcitrance of biofilm-associated infections is mostly due to this increased persistence, which also plays a substantial role in treatment failure (Wainwright *et al.*, 2021; Stojowska Swędrzyńska *et al.*, 2023; Theis *et al.*, 2023).

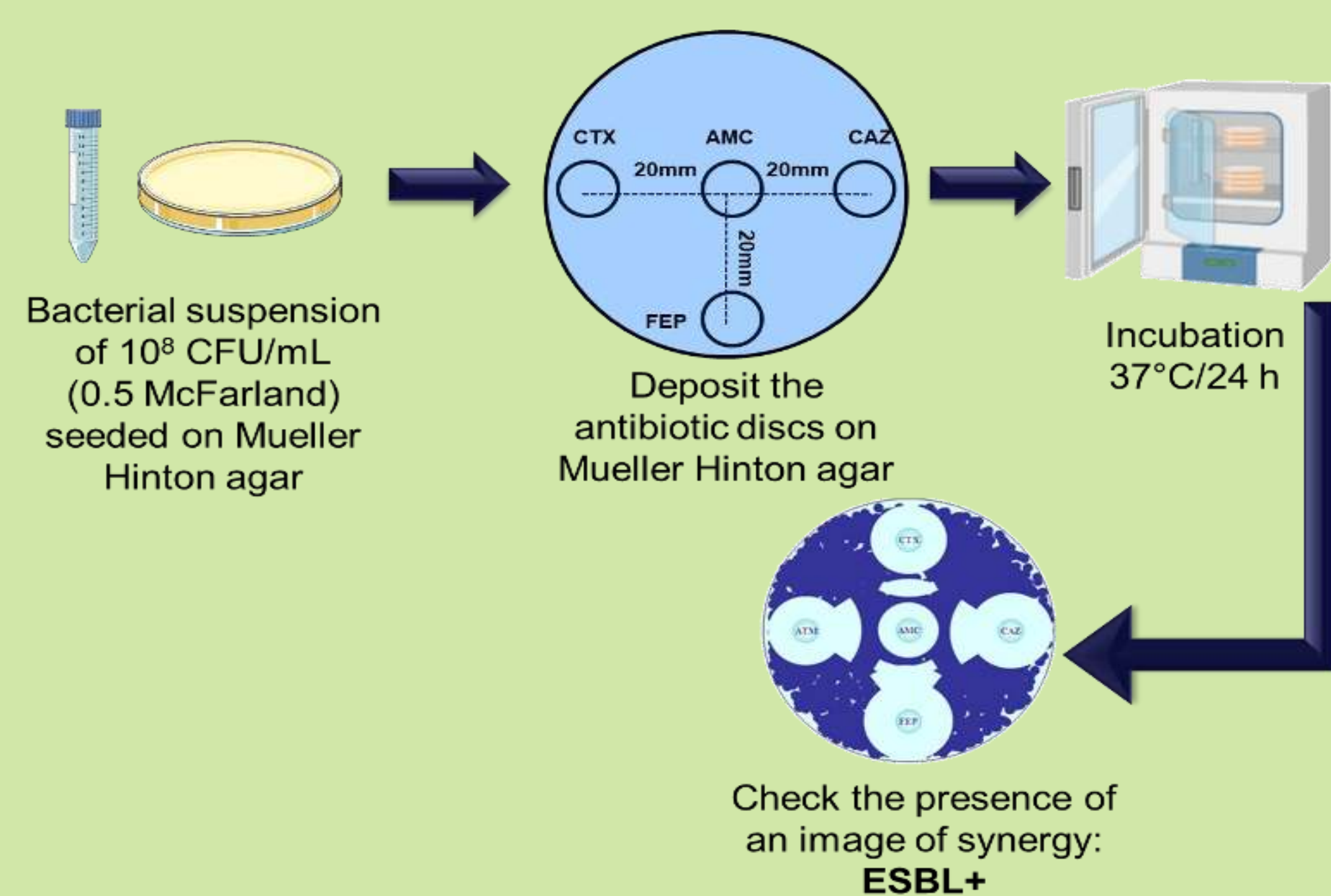
The study's goal is to characterize the antibiotic resistance profiles and pathogenic potential of bacterial strains that isolated from medical devices collected from patients hospitalized at least 48 h in different departments at two hospitals and a private healthcare unit (Bejaia, Algeria).

Materials and methods

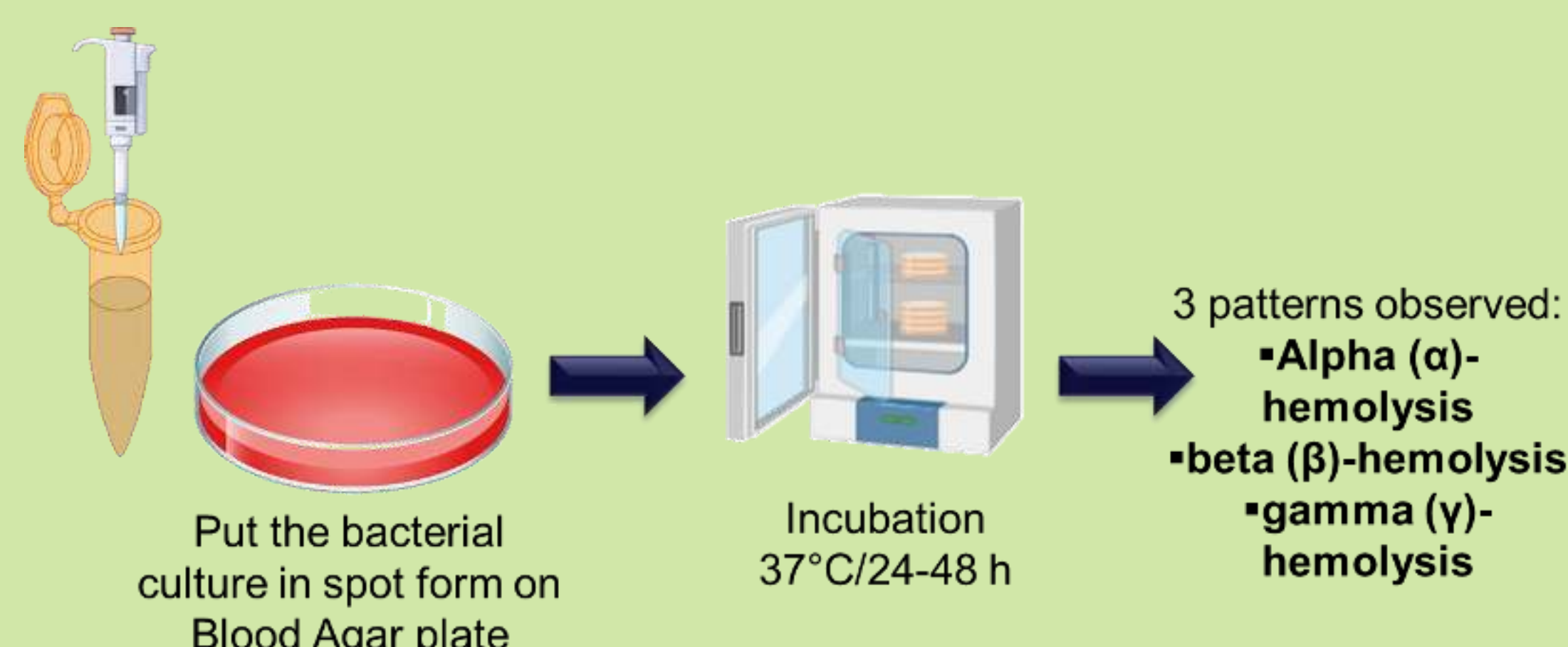
Antibiotic susceptibility testing



Extended Spectrum β -Lactamases (ESBL) production



Hemolytic activity



Results

Antibiotic resistance profiles of the strains

E.coli : NA: 77.42%; TET: 38.71%; AMC: 22.58%

K.pneumoniae: AMC: 100%; FEP: 100%; CAZ,CTX: 83%; TET: 83.33%; GEN: 33.33%

Acinetobacter spp.: FEP, IMP, TET, CIP, COTRI: 100%; CTX, CAZ: 83%

S.aureus: ERY, GEN, TET, CIP: 4.16%

Negative coagulase *Staphylococcus*: ERY: 46.22%; TOB: 42.86%; FOX: 32.77%; CIP: 31.93%; GEN: 31.09%; COTRI: 21.85%; RIF: 7.56%; TET: 6.72%

En.faecalis: TET: 87.88%; ERY, CIP: 81.82%; RIF: 15.15%

Figure 1. Proportion of antibiotic resistance of the pathogenic strains.

Extended Spectrum β -Lactamases (ESBL) production

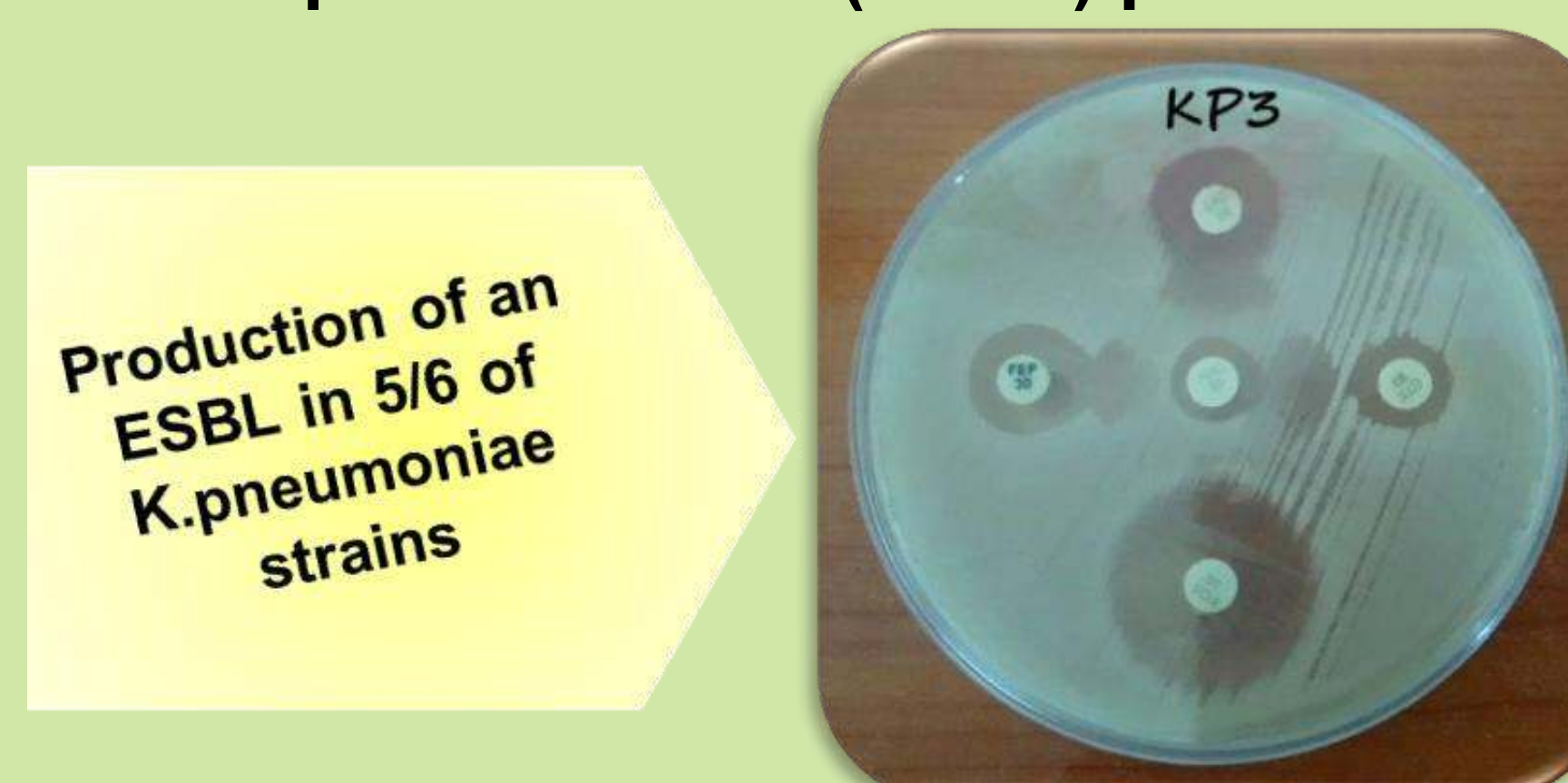


Figure 2. Photography of an ESBL-producing KP3 strains

Hemolytic activity

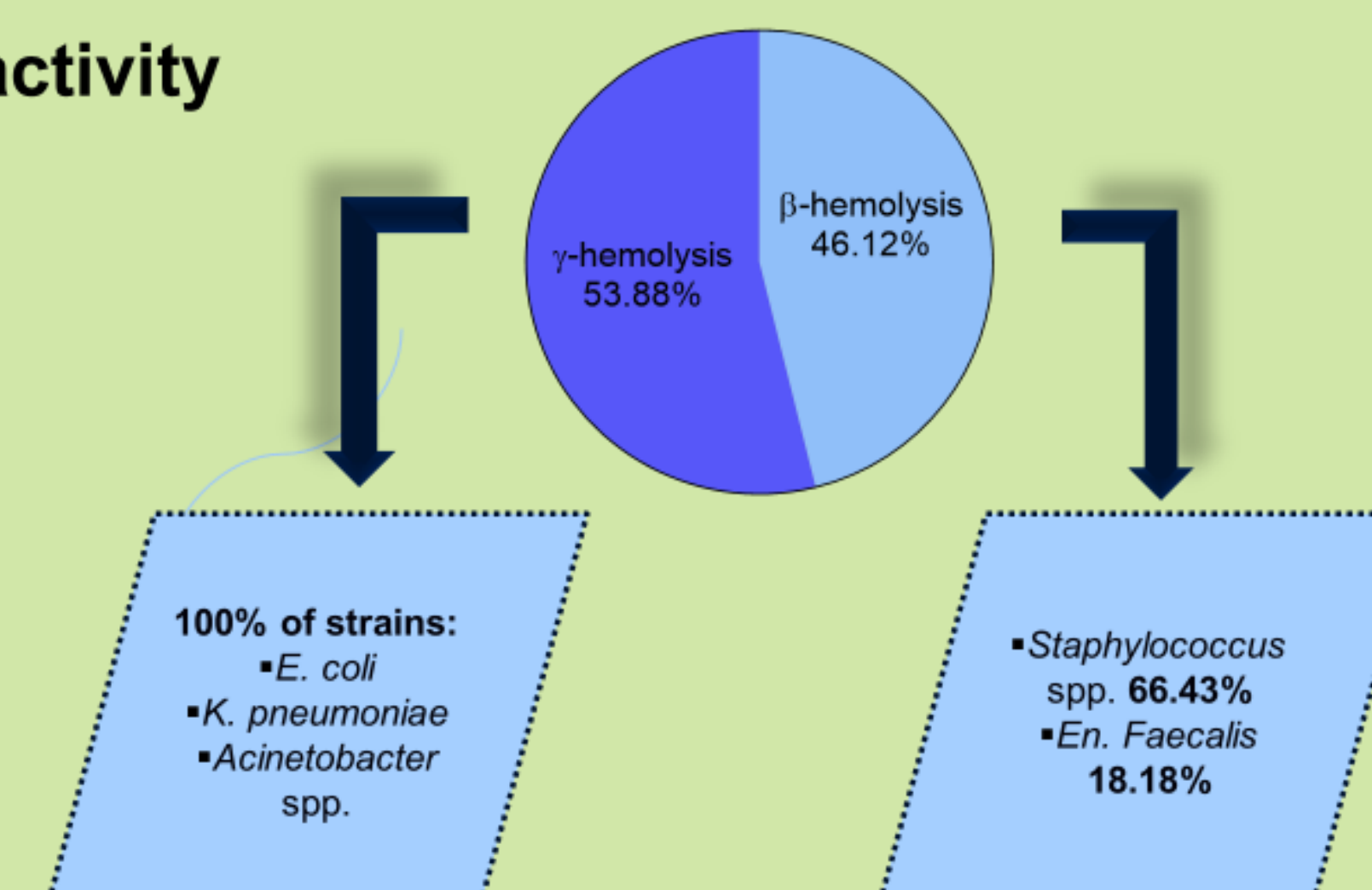


Figure 3. Proportion of Hemolytic activity type of the pathogenic strains.

Conclusion

- Our results showed that virulence traits of these pathogenic strains such as antibiotic resistance and hemolysis factor accentuates their adhesion and their persistence on medical devices and reduce the efficiency of antibiotics.
- Novel antimicrobial agents are urgently needed to treat hospital-acquired infections related to biofilms.

References

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